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Discursive constructions of scientific (Un)certainty about the health risks of China's air pollution: A corpus-assisted discourse study

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ABSTRACT

This article gives a corpus-assisted discourse study of the representations of the health risks of China's air pollution (2011–2014) in Chinese and Anglo-American English-language newspapers with a view to examining their particular ways of constructing scientific (un) certainty about health risks. Findings suggest that although health information has been addressed in the two groups of newspapers, the Anglo-American English-language newspapers prefer to amplify and dramatize the certainty about health risks through such discursive strategies as predication, nomination, and the rhetoric of quantification, while the Chinese English-language newspaper is inclined to highlight the uncertainty about health risks through such discursive strategies as particularizing the Chinese context, complicating the causes of health problems, and arguing for more scientific tests.

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1. Introduction

After thirty years of rapid industrialization, China has now been confronted with the worst air pollution in the world (Yang et al., 2015). The problem has become increasingly severe in the last few years, as can be witnessed from the continual outbreaks of the all-enveloping smog in large areas of China, in particular its capital Beijing. Along with the worsening air pollution come the growing public concerns about its potential health risks (Petts, 2005). As Douglas and Wildavsky (1982, p. 10) suggest, modern individuals are afraid of 'nothing much … except the food they eat, the water they drink, the air they breathe, the land they live on, and the energy they use' (cited in Lupton, 1993, p. 425).

While it has been widely acknowledged that some chronic diseases might be affected by air pollution, such as asthma, allergies, and bronchitis, the exact causal links between air pollution and these diseases remain largely unknown (Petts, 2005). Public understanding of the health risks of air pollution tends to be shaped by not only scientific studies but also, or more importantly, the particular ways these studies are communicated. The mass media is particularly important in informing the public of emerging health risks (Tang and Rundblad, 2015), because health risks 'constantly make headlines in the news media and are increasingly the subject of public communication campaigns' (Lupton, 1993, p. 425). However, risks are not mere objective conditions out there waiting to be represented, but are rather socially constructed (Höijer, 2010; Stallings, 1990). Instead of representing health risks in a neutral way, the media may choose to construct them in their

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preferred way to advance their own socio-political agendas (Collin and Hughes, 2011), so media representations of health risks have been subject to increasing scrutiny in the last two decades (Lupton, 1993).

China's mounting environmental challenges and their communication have already drawn increased attention in academia (Geall, 2011; Ma, 2015; Shapiro, 2012; Tilt and Xiao, 2010; Tong, 2014), but the representations of China's air pollution are still under-examined, with only a few exceptions (S. Chen, 2014; Y. Chen, 2014; Xu, 2012). Even fewer are studies that address the communication of health risks of China's air pollution (Yang et al., 2015), to say nothing of a comparative study of their media representations in different socio-political contexts. Previous studies have demonstrated that the representations of the health risks of a certain issue are apt to vary with different institutions (Solin, 2004; Tang and Rundblad, 2015) and socio-political contexts (Olausson, 2009), but still little knowledge can be gained as regards media constructions of the health risks of China's air pollution in different socio-political contexts. The present study is intended to fill the gap by giving a comparative study of the representations of China's air pollution in Chinese and Anglo-American media, with a particular attention to their particular ways of constructing scientific certainty and uncertainty about its health risks.

2. Scientific (un)certainty in media representations

Uncertainty is one of the main features that characterize many issues in the contemporary world, due to either a lack of knowledge or competing interpretations of data (Friedman, 1999). This is especially visible in the communication of the health risks of some contentious issues, such as climate change, genetic engineering, nuclear power, and air pollution (Olausson, 2009), because their exact health impacts are very difficult to be determined at the present stage. Nevertheless, they are receiving extensive coverage in the media, since 'controversy and debate are important criteria for newsworthy stories' (Friedman, 1999). Media do not merely reproduce how scientists negotiate and express uncertainty, but may represent uncertainties in ways that serve their own ideological purposes. As Campbell (1985) argues, uncertainty itself does not necessarily incur controversy, but serves as a question for negotiation, decision and argument. It is a strategic element of argument rather than something that causes argument. It can be interpreted, negotiated, and constructed in multiple ways (Friedman, 1999).

The discursive constructions of scientific (un)certainty in media, therefore, have been extensively examined in the last few decades (Collins, 1987; Friedman, 1999; Olausson, 2009; Zehr, 2000). While some studies expose that journalists tend to minimize uncertainties in order to create sensational news, some others reveal that they also make scientific claims seem uncertain to serve certain purposes (Stocking, 1999). For example, previous studies on the representations of climate change in the US and the European media have revealed a high degree of uncertainty in the former but plenty of certainty in the latter, which was attributed to the different political stances and measures towards climate change in the US and the European contexts (Boykoff, 2008; Olausson, 2009; Zehr, 2000).

Similar findings have also been made on the discursive constructions of scientific (un)certainty about health risks. In an early study on the British press coverage of food poisoning in the late 1980s, Fowler (1991) suggested that it tended to be characterized by a 'hysterical style', which frequently drew upon the discourses of 'fear', 'hazard', and 'risk'. Brittle and Zint's (2003) examination of the coverage of lead health risks to children in 152 American newspaper articles demonstrated that they deliberately downplayed the certainty of health risks through confining the issue to a local news story and providing less information on health effects. Solin (2004) examined the reproduction of the causality claims between health problems and air pollution in four different genres in the United Kingdom, and demonstrated that news reports tended to strengthen the causality within scientific and government claims. Tang and Rundblad's (2015) recent study on media reporting of health risks of water contaminants in both the UK and the US has also revealed that water organization reports tended to downplay the probability and severity of the risk posed by contaminants, whereas media texts were inclined to highlight their potentiality to be an unconfirmed threat to public health.

3. Corpus-assisted discourse study

The present study combines corpus linguistics (CL) methods (Baker, 2006) and critical discourse analysis (CDA) (Fairclough, 1995; van Dijk, 1998) by adopting the corpus-assisted discourse studies (CADS) approach (Partington, 2004). The benefits of using CL methods in (critical) discourse analysis have been widely acknowledged (Baker et al., 2013). CL methods can enable discourse analysts to process large quantities of data quickly, identify significant linguistic phenomena which might otherwise be missed, reduce the researcher bias by focusing on frequency and saliency, and thus enhance the credibility of analytic results (Baker, 2006).

CL methods have been employed to investigate the meaning of *risk* (Adolphs et al., 2004; Hardy and Colombini, 2011) as well as the particular ways of constructing health risks in different contexts (Adolphs et al., 2004; Brown et al., 2009; Marko, 2010). By analyzing the lemma RISK over the last two decades in the Corpus of Contemporary American English, Hardy and Colombini (2011) have found that 'RISK is genre sensitive, predominantly negative, and heavily medical in shared meaning across genres'. Zinn (2010) goes further to argue for more cross-disciplinary work to connect sociological research interests and corpus linguistic methods to advance our understanding of the risk phenomenon, in particular its cross-cultural differences and social change. Recent attempts can be witnessed in Koteyko et al.'s (2008) investigation of the exploitation of scientific uncertainty about the origin, spread and treatment of MRSA (i.e., methicillin-resistant *Staphylococcus aureus*) in the

media and policy circles of the UK, as well as Tang and Rundblad's (2015) examination of media representations of the health risks of water contaminants in the UK and the US.

Nonetheless, most of these studies fail to give a balanced treatment of the methods and theoretical frameworks traditionally associated with CL and CDA (Baker et al., 2008). Corpus-based studies may adopt a critical perspective but are not well informed by the CDA theory or any particular discourse-oriented theory. Likewise, discourse analysts using CL methods often fail to state their methods explicitly. The CADS approach, however, is characterized by its emphasis on the balanced combination of qualitative discourse analysis and quantitative CL methods, and on the constant movement between the results generated by corpus-analytic tools and the close analysis of linguistic phenomena in their contexts (Haarman and Lombardo, 2009; Morley and Bayley, 2009). This study is thus going to contribute to the growing CADS literature on the discursive constructions of scientific (un)certainty about health risks in news media.

4. Data and methodology

4.1. Data collection and corpus building

One Chinese national English newspaper, i.e., *China Daily* (CD), and three influential Anglo-American English newspapers, i.e., the *New York Times* (NYT) from the US, *The Times* (TT), and *The Guardian* (TG) from Britain, have been selected for investigation. CD is the largest and most influential national English-language newspaper in China. As a typical political party organ, it serves the important functions of external communication and building a positive image for the Chinese government in the international arena. The three Anglo-American English newspapers are selected for a comparative study, owing to their wide international influence and extensive coverage of China's air pollution. Belonging to the same Anglo-American media system, they are known for their ingrained anti-communist ideology (Stone and Xiao, 2007) and their liberal stance in news reporting (Hallin and Mancini, 2012).

The time span set for data collection is from 2011 to 2014. 2011 was the year when the noxious air began to become a topic of day-to-day online and offline conversation in China, while the problem was at its worst in 2014 (van de Ven, 2014). All the news texts concerning China's smog in the four newspapers were collected from the electronic database Lexis-Nexis Academic Universe through the automatic search with the keywords *China* and *smog*. They were further built into two large corpora: CD and AA. The former collects all the news texts in CD, the latter in the three Anglo-American English newspapers. Table 1 provides general information about the two corpora.

Table 1
General information about the CD and AA corpora.

	CD	AA		
		TG	TT	NYT
Number of texts	920	123	104	263
Tokens	506,189	117,224	87,844	242,213
Total	506,189	447,281		

4.2. Methodology

The present study starts from the basic assumption that the two groups of newspapers might diverge in their constructions of health risks of China's air pollution due to their different socio-political contexts. Unlike previous studies devoted to the identification and analysis of certainty and/or uncertainty markers in discourse (Koutsantoni, 2004), the present study does not presuppose any linguistic features that might contribute to the constructions of scientific (un)certainty about health risks in view of the fact that it can be constructed either explicitly or implicitly at different levels of language, such as lexico-grammatical, intertextual, and even thematic levels (Stocking, 1999).

Instead, we propose to analyze the constructions of scientific (un)certainty about health risks from two entry points. The first starts with the words referring to China's air pollution (e.g., *smog*) to examine how the problem is predicated or described. The basic assumption is that scientific (un)certainty about health risks can be detected first through its manner of predication. The second starts with health keywords. Keywords refer to those words which are statistically more frequently used in a corpus than in a secondary or general corpus (Baker, 2006). Health keywords refer to these keywords that are used to discuss health-related information. They are supposed to be crucial to the constructions of health risks in media communication.

With the corpus-analytic tool Wordsmith 5.0, the two corpora were first compared respectively with a general corpus, the 5 million sampler corpus of the British National Corpus (BNC), and two keyword lists were produced. These keywords were ranked in the lists by their keyness values that were calculated by Wordsmith 5.0 automatically based on their log-likelihood. The higher their keyness values are, the more statistically significant these keywords are. Since each list consists of several thousands of keywords, it is beyond the scope of the present study to give a close analysis of all of them. Therefore, only these health keywords among the top 1000 health keywords were examined, their keyness values being above 54.72 in CD and 36.86 in AA, which means that they are all statistically significant. These health keywords were further compared and selected

for contextual analyses in order to examine their contributions to the constructions of scientific (un)certainty about health risks. Since the present study only focuses on health keywords of the two groups of newspapers, we do not take into account the stylistic differences between Chinese, British, and American Englishes, and a pilot analysis has also demonstrated that British and American newspapers do not show significant differences in the choice of these health keywords.

The whole analysis is mutually informed by CL methods and CDA. The corpus-analytic software can help to analyze the large quantity of data, identify statistically significant health keywords, and provide the entry points for further analysis (Baker et al., 2008). The discourse-analytic methods employed in CDA can provide insights for further description, interpretation and explanation of some linguistic features and patterns. A CADS approach can thus help to down-sample the data and provide an illuminating analysis of the discursive patterns that most clearly reveal their respective preferences for health risk constructions.

5. Findings

5.1. Analysis of smog

The (un)certainty of health risks can first be detected through the way China's air pollution, smog in particular, is evaluated. This is called predication strategies, that is, strategies that attribute qualities to smog with the sort of adjectives that are frequently used to describe it. The token *smog* has 759 (0.17%) occurrences in AA and 2477 (0.49%) occurrences in CD. In order to explore how smog is predicated in the two corpora, these collocates at L1 position (the first position to the left) of *smog* are examined, and these collocates with typical evaluative meanings are identified. MI (mutual information) score is used to calculate the strength of the collocation between these evaluative collocates and *smog*, and the minimum frequencies of these collocates are set at three (Baker, 2006). Table 2 shows the top 10 evaluative collocates of *smog* at L1 position. The higher the MI score is, the stronger the collocation is.

Table 2
Top 10 evaluative adjectival and noun modifiers of *smog*.

Rank	CD			AA		
	Collocates	MI score	Freq.	Collocates	MI score	Freq.
1	*Lingering*	13.15331	31	*Oppressive*	8.78407	3
2	*Recurrent*	12.36904	9	*Choking*	7.80318	19
3	*Continuous*	10.78407	3	*Acrid*	7.52104	5
4	*Heavy*	10.76822	362	*Thick*	7.44012	13
5	*Thick*	9.57257	57	*Killer*	7.43358	5
6	*Severe*	9.22465	57	*Murky*	7.19911	3
7	*Worsening*	8.95810	11	*Toxic*	6.59707	14
8	*Notorious*	8.81209	13	*Dense*	6.53615	3
9	*Worst*	8.66710	13	*Heavy*	6.43830	18
10	*Choking*	8.64272	34	*Noxious*	6.26051	3

In CD, the majority of these evaluative collocates underline the severity of China's smog, including *heavy*, *thick*, and *severe*. In particular, the collocate *heavy* occurs much more frequently with *smog* than other evaluative collocates at L1 position. These evaluative collocates are usually found occurring together with other weather phenomena such as rain, snow, and storm. They help to construct smog as a kind of weather condition, to the neglect of its particular health impact. Some other evaluative collocates also highlight the deteriorating and unexpected nature of smog, such as *worsening*, *notorious*, *worst*, *recurrent*, *continuous* and *lingering*, but they have nothing to do with its health impact. Only the collocate *choking* in CD suggests the uncomfortable physical reaction to smog. In contrast, apart from highlighting the severity of China's smog (such as *thick*, *heavy*, and *dense*), AA tends to dramatize the health risks of China's smog, as can be witnessed from these collocates like *oppressive*, *choking*, *acrid*, *killer*, *murky*, *toxic*, and *noxious*. While *oppressive* and *choking* indicate the uncomfortable physical reaction brought by smog, collocates such as *acrid*, *toxic*, *noxious*, and *killer* amplify the poisonous and lethal nature of smog. The differences in the choice of evaluative adjectives actually shed light on their different views of the certainty and uncertainty of China's smog: AA tends to indicate the certainty of the health risks of China's smog by describing health risks as its inherent attribute, while CD prefers to construct China's smog as a weather phenomenon, suggesting its view of the uncertainty of the health risks of China's smog.

5.2. Analysis of health keywords

Table 3 shows the health keywords identified in the two corpora. One interesting finding is that an equal number of health keywords can be identified in both corpora. This can be attributed to the fact that one of the primary public concerns towards air pollution is its health impact. Several health keywords can be identified in both corpora, such as *health*, *hazardous*, *lung*, *lungs*, *respiratory*, *choking*, *cancer*, *unhealthy*, *asthma*, and *choked*. Among them, *health* is the most key keyword in both

Table 3
Health keywords of CD and AA.

Rank	AA				CD			
	Rank1[a]	Tokens	Freq.	Keyness	Rank2[a]	Tokens	Freq.	Keyness
1	61	HEALTH	402	418.1	124	HEALTH	482	525.5
2	101	HAZARDOUS	90	290.5	219	HAZARDOUS	105	331.3
3	105	LUNG	75	286.3	233	LUNGS	89	307.6
4	110	LUNGS	77	274.1	251	LUNG	78	282.7
5	118	DEATHS	93	257.2	279	RESPIRATORY	67	260.2
6	130	TOXIC	85	225	311	CHOKING	62	227.1
7	149	RESPIRATORY	52	205.7	348	CANCER	121	203.2
8	169	CHOKING	50	186.6	429	DISEASES	64	152
9	176	AIRPOCALYPSE	37	181.6	494	HEALTHY	93	126.8
10	206	CANCER	99	160.2	534	FITNESS	49	116.5
11	222	UNHEALTHY	44	151.7	552	BREATHE	48	111.6
12	416	ASTHMA	30	86.9	639	UNHEALTHY	32	94
13	418	BREATHING	49	85.9	656	HARMFUL	49	91.6
14	427	NOXIOUS	23	84.4	746	CHOKED	25	77.2
15	430	DANGEROUS	82	83.9	858	AGING	19	65.4
16	580	DIED	77	62.7	898	HAZARD	34	61.2
17	979	CHOKED	14	37.9	999	ASTHMA	23	55.1

[a] The actual rank of these health keywords in the top 1000 keywords.

corpora, suggesting that both corpora are keen on talking about health-related issues. *Hazardous* and *unhealthy* indicate that the harmful nature of China's smog or air pollution has been mentioned in both corpora. *Lung* and *lungs* emerge as keywords, because lungs are deemed most prone to be affected by air pollution. *Respiratory*, *cancer*, and *asthma* refer to the problems and diseases that can be brought by smog, such as respiratory problems, lung cancer, and asthma. *Choking* and *choked* are highly valued in both corpora, because they are the most direct physical response to smog. Therefore, the health impact of Chinese smog, as a theme, has been addressed in both corpora.

Nevertheless, the two corpora also show their respective preferences for certain health keywords. AA prefers such keywords as *deaths*, *toxic*, *airpocalypse*, *breathing*, *noxious*, *dangerous*, and *died*. Among them, *toxic*, *noxious*, and *dangerous* point to the poisonous nature of smog, and *deaths* and *died* suggest the lethal nature of smog. The keyword *airpocalypse* even highlights China's air pollution as an apocalyptic disaster, implying that nobody in China can escape from it. On the contrary, CD tends to present a moderate picture of the health risks of China's smog with some keywords, such as *hazard*, *harmful*, *disease*s, *aging*, *fitness*, and *healthy*. While *hazard*, *harmful*, and *disease*s may suggest the potential health risks of smog, *aging*, *fitness*, and *healthy*, in fact, have nothing to do with the health risks of China's smog, because they refer primarily to China's aging population, fitness campaign, and healthy lifestyle. Therefore, AA seems to present a more severe picture of the health risks of China's air pollution than CD, even though the health theme has been addressed in both of them. In order to further test our hypothesis, the following sections give a close analysis of three keywords in their specific contexts of use, i.e., *health*, *lung*, and *deaths*. They are believed to address three key public concerns: (1) whether China's air pollution has health impacts; (2) how air pollution affects the body; and (3) how many deaths air pollution can cause.

5.2.1. Health

Health is the most key as well as the most frequently used health keyword in both corpora. An examination of its collocates finds that *pollution* ranks among its top 10 most frequent lexical collocates in both corpora (27 in AA and 28 in CD), so the collocation of *health* and *pollution* is examined in its contexts. According to Solin (2004), while there may be a general agreement that air pollution is an unwanted thing, disputes remain as regards the severity of the problem.

Their divergent ways of constructing health risks can first be detected through their preferences for naming the health impacts of air pollution, i.e., nomination strategy. AA prefers to designate health effects by using such nominal groups as *China's biggest* ***health*** *threat*, *the biggest public* ***health*** *risk*, *the huge* ***health*** *cost of urban pollution*, *pollution-linked* ***health*** *problems*, and *a* ***health*** *hazard*. In these nominal groups, the head (such as *risk*, *hazard*, *problem*, *hazard*, and *cost*) carries the implicature that pollution can cause health problems. These nominal groups help construct the health problems of air pollution as presupposed, background information. On the one hand, it makes them hardly contestable, and, on the other hand, it creates a consensus concerning the health risks of air pollution and takes any discussion or debate about them off the table (Dunmire, 2005). Besides, some of the heads of these nominal groups are further modified by amplifying evaluative words like *biggest* and *huge*, which help to augment the certainty of health risks of China's air pollution. These nominal groups, therefore, help to transform the possible and potential damages into known and inevitable damages, and elevate the discussions about whether air pollution can cause health problems to the discussions about how severe the health impacts of air pollution are. By contrast, CD prefers to use nominal groups with neutral words as the head, such as *the effects of air pollution*, *the influence of air pollution*, and *the impact of air pollution*. They suggest CD's reluctance to label the exact health impacts of air pollution, thus contributing to scientific uncertainty about its health risks.

The certainty about health risks in AA has also been constructed through some (40 occurrences) explicit statements and assertions, either by some experts or by news writers themselves, as in the following:

(1) Birkett says, exposure to air *pollution* is the biggest public *health* risk after smoking in the UK; government figures suggest 29,000 deaths a year could be attributed to it.
(TG, 2012/08/06)

(2) "Air *pollution* is clearly linked to *health*, and the great thing is that everybody—that's government officials and company executives alike—breathes the same air."
(NYT, 2013/09/01)

These statements make an absolute link between air pollution and health risks. The simple present tense they use contributes to the construction of these statements as a matter of fact, allowing no space for negotiation and contestation. The constructions of scientific certainty about health risks are especially visible in the use of the certainty marker *clearly* in Example (6). By using the word *clearly*, the speaker implies that the same inferences can be made between the speaker/writer and his listeners/readers based on their shared knowledge, or they are missing the obvious (Hyland, 1998; Koutsantoni, 2004). It helps to rule out possible alternatives and promote the views as generally held and consensual. It follows that the exact links between health risks and air pollution are rarely questioned in the use of *health* in AA.

While CD acknowledges that air pollution can cause health problems, it refuses to make the explicit linkage between air pollution and health risks. Even in the statements about the potential health problems of air pollution, they are often hedged by modal verbs or conditions, suggesting the uncertainty about the health risks of air pollution, as in the following:

(2) "It's true that air *pollution* can harm human health, but to what extent is still being researched.
(CD, 2013/11/11)

(4) In December, Chen Zhu, former minister of health, wrote in an opinion piece in the British general medical journal The Lancet that a 2010 study finding that at least 350,000 people in China die prematurely each year as a result of outdoor air *pollution* "might overestimate the *health* effects of outdoor air pollution".
(CD, 2014/01/08)

As the above examples show, modal verbs, such as *might*, *can*, and *could*, are frequently used to hedge the statements about the health impacts of air pollution, thus downplaying their certainty (Hyland, 1998). Besides, CD tends to use a *concede* + *counter* pairing (Martin and White, 2005). While acknowledging the potential health impacts of air pollution, it immediately presents an opposing view that mitigates the certainty of the previous claims. The pitting of one expert's or group's views against another's can contribute to the constructions of the uncertainty about the health risks of China's air pollution (Friedman, 1999).

Besides, CD tries to construct the uncertainty about the health risks of China's air pollution through particularizing the Chinese context. It frequently (44 occurrences) emphasizes the absence of relevant studies on the health impacts of air pollution in the Chinese context (see Example 5), and underlines the necessity for more time, more data, and more tests in the Chinese context to determine the exact health impacts of air pollution (see Examples 6 and 7).

(5) Yu Fang, a researcher at the Chinese Academy for Environmental Planning, who was part of a 2013 study on outdoor pollution and health in China, said current research on the *health* effects of long-term exposure to air *pollution* can only refer to similar studies in the United States, due to a lack of relevant studies in China.
(CD, 2014/01/08)

(6) Wang Wuyi, an environmental and health researcher at the Chinese Academy of Sciences, said *more data is needed* on the effects of air *pollution* on people's ***health***.
(CD, 2014/01/08)

(7) Mao Qun'an, spokesman for the National Health and Family Planning Commission, said in a news conference on Tuesday that the authority's monitoring of the ***health*** effects of outdoor air *pollution* could yield results *only after a long period*.
(CD, 2014/04/09)

One typical strategy of constructing claims to truth in scientific discourse is to remove contingencies from the statements (Zehr, 1999). These contingencies might suggest the specific technical, political, or social conditions of these statements. The elimination of these contingencies thus contributes to the certainty of these statements, as AA does in its representations of the health impacts of China's air pollution. The accentuation on these contingencies, on the contrary, can magnify the uncertainty of these claims (Friedman, 1999), and this is what CD does in its representations of the health risks of China's air pollution.

5.2.2. Lung

The keyword *lung* refers primarily to lung cancer, because most of its occurrences (50 out of 78 in CD, and 44 out of 75 in AA) collocate with *cancer* in both corpora. A close examination of the collocation of *lung* and *cancer* in AA reveals two main types of discourses surrounding *lung* and *cancer*. The first type (29 occurrences) emphasizes the severity of lung cancer in

China. It emphasizes that lung cancer has increased rapidly and grown to be a leading form of cancer in China. It has been characterized by a high degree of certainty, because these statements are rarely hedged, as in the following:

(8) The report also said *lung* cancer was the leading form of cancer in China.
(NYT, 2013/11/05)

(9) The *China Daily*, for example, recently reported that the *lung cancer* rate has increased by 60 percent in Beijing over the last decade, though the smoking rate hasn't changed.
(NYT, 2011/12/09)

While the emphasis on the dramatic increase of lung cancer implicates the negative health impact of air pollution, the other type of discourse (9 occurrences) addresses air pollution's contributions to lung cancer. A strong causation has been established through the use of causative verbs (e.g., *cause*), verbs expressing result (e.g., *linked to*, *attribute … to*), and adverbials expressing cause/result (e.g., *as a result of*) (Solin, 2004), as in the following:

(10) In October the World Health Organization' s International Agency for Research on Cancer (IARC) classified air pollution as carcinogenic. It said that there is "sufficient evidence" that expose to air pollution causes *lung cancer* and that exposure has increased significantly in "rapidly industrial countries with large populations". (TG, 2013/10/24)

(11) Zhong Nanshan, a respiratory expert at the Chinese Academy of Engineering, told China Daily last month that without intervention, PM 2.5 particulates would replace smoking as China's top cause of *lung cancer*. (NYT, 2012/01/28)

The tendency is most revealing in its representations of an 8-year-old girl who was diagnosed with lung cancer. AA shows a strong interest in the coverage of this incident, since it has given an extensive coverage of the story. Besides, it is eager to put the blame on air pollution, even though no scientific evidence has been employed to support the claim. Instead, it attributes the responsibility for the claim to the girl's doctor, as in the following:

(12) But nothing is quite so shocking as an 8-year-old girl being diagnosed with *lung cancer* which her doctor says is a result of air pollution. (TG, 2013/11/07)

As Seale (2002) suggests, media health stories are keen on dramatizing the vulnerability of victims through reference to their smallness, helplessness, or innocence in order to attract human interest. A terminally ill child with a rare form of cancer, therefore, carries high news value. The goodness and innocence of the child is contrasted with the sheer malignance of air pollution, thus contributing to the generation of health scare stories, which explicitly attribute serious health risks and various other sorts of harm to air pollution.

Although the same two types of discourses can be identified in CD, they are constructed in different ways. CD also acknowledges the rise of lung cancer in China, but it prefers to use technical terms to mystify lung cancer. A distinction has been made between two types of lung cancer: 'lung adenocarcinoma' and 'squamous cell lung cancer'. It is reported that while the proportion of the former increased, the proportion of the latter decreased, as in the following:

(13) "Of the city's lung cancer cases that were histologically diagnosed, the proportion of squamous cell lung cancer decreased yearly from 30.41 percent in 1998 to 24.16 percent. Meanwhile, the proportion of lung adenocarcinoma increased from 42.83 percent to 46.80 percent," the study found.
(CD, 2014/02/27)

Instead of attributing lung cancer directly to air pollution, CD highlights the multifarious factors that may contribute to lung cancer, thus making its causes much more complicated, as in the following:

(14) Feng, the doctor, said in an e-mail to China Daily: "Many factors contribute to lung cancer, such as smoking, ionizing radiation, genetic factors and air pollution …"
(CD, 2013/11/11)

Likewise, although CD also represents the 8-year-old girl who was diagnosed with lung cancer, it refuses to attribute the girl's lung cancer immediately to air pollution. It emphasizes the difficulty in identifying the causes for the girl's lung cancer (see Example 15), and denies the statement made by the girl's doctor in AA (see Example 16).

(15) The cause of her lung cancer could not be determined because it was complicated to identify, the hospital wrote in an e-mail to China Daily at the weekend.

(CD, 2013/11/10)

(16) A cancer hospital in Jiangsu province has denied media reports that an 8-year-old girl became China's youngest lung cancer patient due to worsening air pollution.

(CD, 2013/11/11)

In summary, although both CD and AA have covered lung cancer, AA tends to highlight its dramatic increase in China and identify air pollution as the main culprit, whereas CD makes a distinction between two types of lung cancer, and refuses to attribute them directly to air pollution. In their representations of the 8-year-old girl with lung cancer, AA is anxious to put the blame on air pollution, while CD argues for the uncertainty of its causes. Therefore, AA prefers to construct the certainty about air pollution's contributions to lung cancer, while CD tends to construct an unresolved relationship between them.

5.2.3. Deaths

Although *deaths* ranks among the top 1000 health keywords in AA but not in CD, it is examined because it suggests the lethal consequences of smog. AA's emphasis on deaths can be viewed as a typical health scare strategy. A contextual analysis of 93 occurrences of *deaths* reveals that 77 of them emphasize the death toll that has been and will be caused by air pollution or smog, as in the following:

(17) According to Gray's study, while the growth of coal consumption has slowed, 570 new coal-fired plants are either being built or are planned, and if they go ahead would be **responsible for** a further 32,000 premature *deaths* each year. (TG, 2013/12/12)

(18) China's coal emissions **responsible for** 'quarter of a million premature *deaths*'

(TG, 2013/12/12)

Several discursive patterns can be identified in AA's representations of deaths caused by air pollution. First, the token *deaths* has been frequently modified by the adjective *premature*, emphasizing that they are not natural deaths, but caused by something else. In the present context, AA emphasizes that this kind of deaths is caused by China's air pollution, because it explicitly relates deaths to air pollution with such expressions as *contribute to*, *result in*, *responsible for*, *involved*, etc. Besides, these expressions are rarely hedged, and are thus presented as a matter of fact or as the findings of some scientific studies or reports. The reliability of these claims is seldom questioned. Furthermore, *premature deaths* is often quantified by large accurate numbers, such as *up to 2.6 to 4.4 million*, *1.2 million*, *32,000*, etc. These particular ways of quantification represent 'the rhetoric of quantification', because they cannot be understood as the literal rendition of the facts about the exact number of deaths, nor does it mean that they are purely wrong (Potter et al., 1991).

The use of large numbers serves two primary ideological functions here. On the one hand, it can dramatize the health risks of China's air pollution by demonstrating the magnitude of the disaster (Collin and Hughes, 2011), and these large numbers represent the so-called 'scare statistics', so it is a resource often exploited to argue for the severity of social and health problems (Petersen and Deborah, 1996). The immense symbolic power it holds can help to intensify hysteria and create an alarming impact on public consciousness (de Souza, 2007; Fowler, 1991). On the other hand, these accurate numbers also help to create the impression of objectivity and credibility (Bell, 1991), and represent an acceptable rhetorical strategy for serious newspapers (Collin and Hughes, 2011). The citation of these figures from prominent news sources and news actors also increases the credibility of these numbers (Lupton, 1993). They contribute to the construction of China's air pollution as an imminent and immense health disaster. The frequent use of quantification thus not only implies certainty but also invokes 'airpocalyptic' imagery.

An examination of the 36 occurrences of deaths in CD reveals that the accuracy of the number of premature deaths is frequently questioned. CD argues that in order to arrive at an accurate number, a domestic study on a large sample of the population with a long life span is required. More studies are required because many factors may affect the results of scientific studies. Therefore, CD tends to emphasize the uncertainty about the number of deaths caused by smog, as in the following:

(19) "To get the most-accurate number of premature *deaths* caused by air pollution in China, a domestic cohort study covering a large-enough sample of the population and a long life span is a precondition," said Kan. (CD, 2014/02/26)

(20) The World Health Organization recently named air pollution, and especially PM2.5, as a leading environmental cause of cancer *deaths*. Chen says more tests need to be done to answer the small particle question. (CD, 2014/02/26)

6. Conclusion

To sum up, while the two groups of newspapers address health risks as a key theme in their representations of China's air pollution, they show distinct differences in their constructions of the (un)certainty about these risks. Anglo-American English newspapers tend to construct these risks with absolute certainty through a number of discursive strategies, such as

predication, nomination, the rhetoric of quantification, and explicit statements. CD prefers to emphasize the uncertainty about the exact health risks of China's air pollution, even though it acknowledges that it may cause health problems. It is achieved through particularizing the Chinese context, complicating and mystifying the connections between health problems and air pollution, and arguing for more tests, more data, and more time to determine the exact health risks of air pollution. While both groups of newspapers represent China's air pollution as a scientific issue and resort to experts and scientific studies for its representations, they tend to present competing pictures about its health risks.

The systematic differences in their constructions of health risks of China's air pollution, however, cannot be properly understood either in scientific terms or in terms of the objectivity of news reporting. It is improper and unnecessary to judge whose representations are closer to scientific truths or more objective, because the answers to these questions, like the issue itself, cannot be determined right away as regards the inherent uncertainty of scientific findings. As Douglas and Wildavsky (1982) argue, the risks that are selected may have nothing to do with the real danger but are subject to the influences of socio-cultural factors. Like many other Chinese problems, such as human rights, trade disputes, and national crises, China's air pollution receives extensive media attention from Anglo-American English newspapers, because it can contribute to constructions of the negative image of a failing Chinese government unable to cope with its own problems. This is consistent with the ingrained anti-communist ideology that characterizes the dominant Anglo-American English media (Herman and Chomsky, 1988; Stone and Xiao, 2007). The all-enveloping smog in China thus provides a good opportunity for them to depict an apocalyptic China, so they spare no efforts in amplifying and dramatizing the certainty about its health risks.

As a typical party organ, CD takes responsibility for communicating voices of the Chinese government to the international community and building a positive image of China. The ever more prevalent public concerns regarding China's air pollution and its potential health risks make it impossible for the Chinese government to keep silent on the issue. CD's representations of the health risks of China's air pollution, therefore, can be better viewed as essentially dialogic in nature. It helps to argue against the certainty claims from the dominant Anglo-American news media and presents an uncertainty picture about the health risks of China's air pollution. That may explain why similar health discourses can be identified in CD and Anglo-American English newspapers, though constructed in different or competing ways. Moreover, the constructions of scientific uncertainty about the exact health risks of China's air pollution point to the necessity for more scientific research in the future. They can help to legitimate the status quo (Zehr, 2000) and win more time for the Chinese government to control air pollution. This is equally important in view of the fact that air pollution is an unwanted product of China's industrialization, so it cannot be brought under immediate control without sacrificing China's economic development.

The present study, therefore, has revealed not only the competing ways of constructing the certainty and uncertainty of the health risks of China's smog but, more importantly, the socio-political motivations behind their particular ways of constructions. Since the exact causal links between air pollution and health problems remain largely unknown (Petts, 2005), the preferential constructions of the certainty and uncertainty of health risks in news media might serve important social and ideological functions. While media representations of health risks have drawn the interest of several disciplines, such as risk psychology, sociology of risk, and media studies (Tulloch and Zinn, 2011), a CADS approach can bring together the insights from both micro linguistic analyses and macro socio-political analyses, exposing the internal inconsistences and contradictions of media representations of health risks, and the opaque as well as transparent relationships between health risk discourse and socio-political contexts (Tang and Rundblad, 2015). It can thus provide a better understanding of the discursive constructions of health risks by media from different socio-political contexts, and contribute to the growing interdisciplinary research on health, risk, and the media (Tulloch and Zinn, 2011).

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